

Total No. of Questions : 4]

SEAT No. :

PD-322

[Total No. of Pages : 2

[6411]-101

B.E. (Mechanical) (Insem.)

QUALITY & RELIABILITY ENGINEERING

(2019 Pattern) (Semester - VIII) (402050 A) (Elective - V)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) All questions are compulsory.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Explain ISO 9000 Quality System Standards with its types. [7]

b) Write a short note on (Any 2) [8]

- i) Quality Circle
- ii) Kaizen
- iii) 5S
- iv) Poka-Yoke

OR

Q2) a) Explain Juran's Trilogy approach with diagram. [7]

b) What is the Cost of Quality? Explain cost of failure, cost of appraisal, and cost of prevention. [8]

Q3) a) What are advantages of sampling inspection over 100% inspections? Explain the difference between single sampling and double sampling plan. [7]

b) A subgroup of 4 items is taken from the manufactured items. After 20 subgroups the values of $\bar{X}$ and R were found out to be $\sum \bar{X} = 350$ and $\sum R = 8.5$. The specification limits of the part are 17.5 ± 0.23 . Assuming the process is in control, what conclusion would you draw about the ability of the process to produce the items within specified limits? Assume for subgroup of 4, $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.283$, $D_2 = 2.059$, $n = 4$, $N = 20$. [8]

OR

P.T.O.

- Q4) a)** Explain in detail: Operating Characteristic Curve showing Producer's Risk, Consumer's Risk, AQL, LTPD, Indifference Region. [7]
- b)** The given table shows the number of defectives found in inspection of 10 lots of 100 items each. Determine the control limits for appropriate chart and state whether the process is in control or not. Draw the control chart. [8]

Lot No.	1	2	3	4	5	6	7	8	9	10
No. of Defectives	6	3	1	4	3	0	5	5	2	3

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